

GCP Architecture & IoT Integration Study

Exploring Functional Concepts, Multi-layered Architecture, and Advanced
IoT Services on Google Cloud Platform

SEMINAR AGENDA



Part 1

Public Cloud Fundamentals & Core GCP Characteristics.



Part 2

GCP Functional Architecture: Compute, Storage, Network, Management.



Part 3

Detailed IoT Focus: Pub/Sub, Dataflow, BigQuery, & Vertex AI.



Part 4

IoT Reference Architectures and Real-world Use Cases.



Part 5

Security, Challenges, and Future Scope (Edge AI).

CLOUD COMPUTING BASICS (NIST MODEL)

Core Essentials

- ✓ **On-demand Self-service:** Provisioning resources without human intervention.
- ✓ **Broad Network Access:** Accessible via mobile, tablet, or desktop.
- ✓ **Resource Pooling:** Multi-tenant model serving multiple consumers.

Deployment Models

While hybrid and private exist, our focus is the **Public Cloud**: Infrastructure owned by a service provider (GCP) and shared by multiple organizations.

GCP CHARACTERISTICS: SCALABILITY

Scalability is the system's ability to handle growing amounts of work or its potential to be enlarged to accommodate that growth.

Auto

SCALING

MIGs

INSTANCE GROUPS

Global

FOOTPRINT

Vertical Scaling: Increasing VM size (CPU/RAM).

Horizontal Scaling: Adding more VM instances to handle traffic spikes, critical for IoT sensors during event bursts.



Scalability in GCP is handled through Managed Instance Groups. In an IoT context, if 1 million sensors suddenly start reporting, the GCP auto-scaler adds nodes instantly.

ELASTICITY & MULTI-TENANCY



Rapid Elasticity

The ability to scale resources back down during low-usage periods to minimize costs. Pay for what you use.



Multi-tenancy

Physical hardware is shared, but logical isolation is maintained through software layers and VPC security.



Availability

99.99% SLAs through regional and zonal distribution of resources.

GCP SERVICE MODELS COMPARISON

Feature	IaaS (Compute Engine)	PaaS (App Engine)	SaaS (Google Workspace)
Control	Maximum (OS level)	Moderate (Code level)	Minimum (App level)
Responsibility	Customer manages OS/Apps	Google manages OS	Google manages everything
Examples	Virtual Machines, Storage	Cloud Functions, App Engine	Drive, Gmail, BigQuery
IoT Fit	Heavy processing nodes	Event-driven logic	Analytics dashboards

FUNCTIONAL ARCHITECTURE: THE 4 PILLARS



Compute Layer

Execution environments (VMs, Containers, Serverless).



Storage & Database

Data persistence (Object, Relational, NoSQL).



Networking

Communication fabric (VPC, Load Balancing, DNS).



Management & Ops

Governance (IAM, Monitoring, Logging, Security).

DEEP DIVE: COMPUTE LAYER

Compute Engine

Virtual Machines on Google's infrastructure. Full control over OS and hardware specs.

Kubernetes (GKE)

Orchestration for containerized microservices. The industry standard for portable workloads.

Cloud Run

Fully managed serverless container execution. Scales to zero when not in use.

Note: IoT edge gateways often sync with GKE clusters for local processing.

DEEP DIVE: STORAGE & DATABASES



Cloud Storage

Object storage for unstructured data. Perfect for IoT logs and firmware images.



Cloud Bigtable

High-performance NoSQL for time-series data. **The primary choice for heavy IoT sensor data.**



Cloud SQL

Fully managed MySQL, PostgreSQL, and SQL Server for relational application data.

DEEP DIVE: NETWORKING LAYER



- ✓ **Global VPC:** Virtual Private Cloud that spans regions without requiring the public internet.
- ✓ **Cloud Load Balancing:** Distributes traffic globally via a single anycast IP.
- ✓ **Cloud Interconnect:** Connects on-premise gateways directly to GCP fiber.

MANAGEMENT, MONITORING & IAM

Cloud IAM

Identity & Access Management. Define "Who" can do "What" on "Which" resource. Critical for device security.

Cloud Operations

Integrated Monitoring, Logging, and Error Reporting. Visualizes the health of your entire cloud environment.

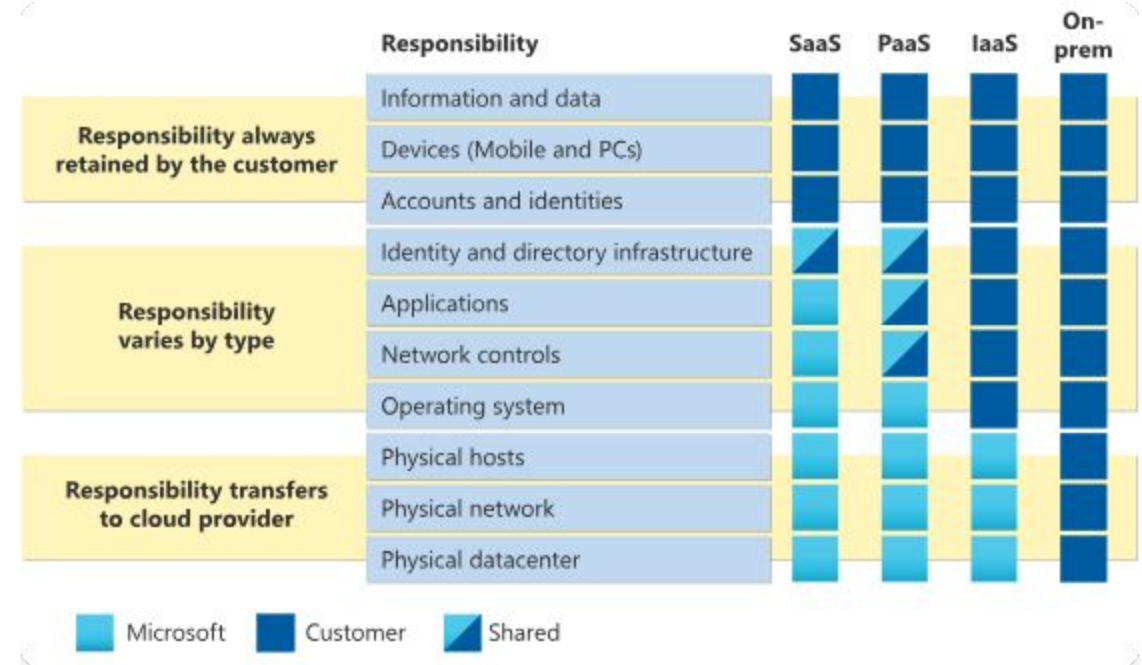
Deployment Manager

Infrastructure as Code. Automate the creation of entire IoT environments via templates.

SECURITY & SHARED RESPONSIBILITY

GCP uses a **Shared Responsibility Model**. Google secures the infrastructure; the customer secures the data and apps.

- ✓ **Google Secures:** Hardware, Network, Physical Security.
- ✓ **You Secure:** Device Identities, Data Encryption, Firewall Rules.



Core Section: IoT Emphasis

Deep Dive into Services, Data Pipelines, and AI
Integration

GCP IOT SERVICE: PUB/SUB

What it is

A global messaging bus that decouples senders (devices) from receivers (analytics apps).

Role in IoT

- ✓ Ingests millions of messages per second.
- ✓ Supports asynchronous communication.
- ✓ Ensures zero data loss during traffic spikes.

GCP IOT SERVICE: DATAFLOW

The Power of Streams

Dataflow provides fully managed, serverless **Streaming** and **Batch processing** based on Apache Beam.

IoT Transformation

- ✓ Filters out "noise" (redundant sensor data).
- ✓ Aggregates data in real-time (e.g., average temp per minute).
- ✓ Enriches data with geographical metadata.

GCP IOT SERVICE: BIGQUERY



Serverless Data Warehouse

Store petabytes of historical IoT data without managing a single server. Use SQL to gain business insights.



Performance

Analyze billions of rows in seconds. Integrated ML (BigQuery ML) allows predictive analysis directly in SQL.

CLOUD FUNCTIONS & AUTOMATION

Event-Driven Logic

Cloud Functions are small pieces of code that trigger automatically based on IoT events.

Use Cases

- ✓ **Alerts:** Send a notification if temperature exceeds 100°C.
- ✓ **Device Control:** Command a fan to turn on if data hits a threshold.
- ✓ **Database Updates:** Update device status in Cloud Firestore.

AI/ML INTEGRATION (VERTEX AI)



Predictive Maintenance

Use sensor history to train models that predict when a machine is likely to fail.



Vision AI

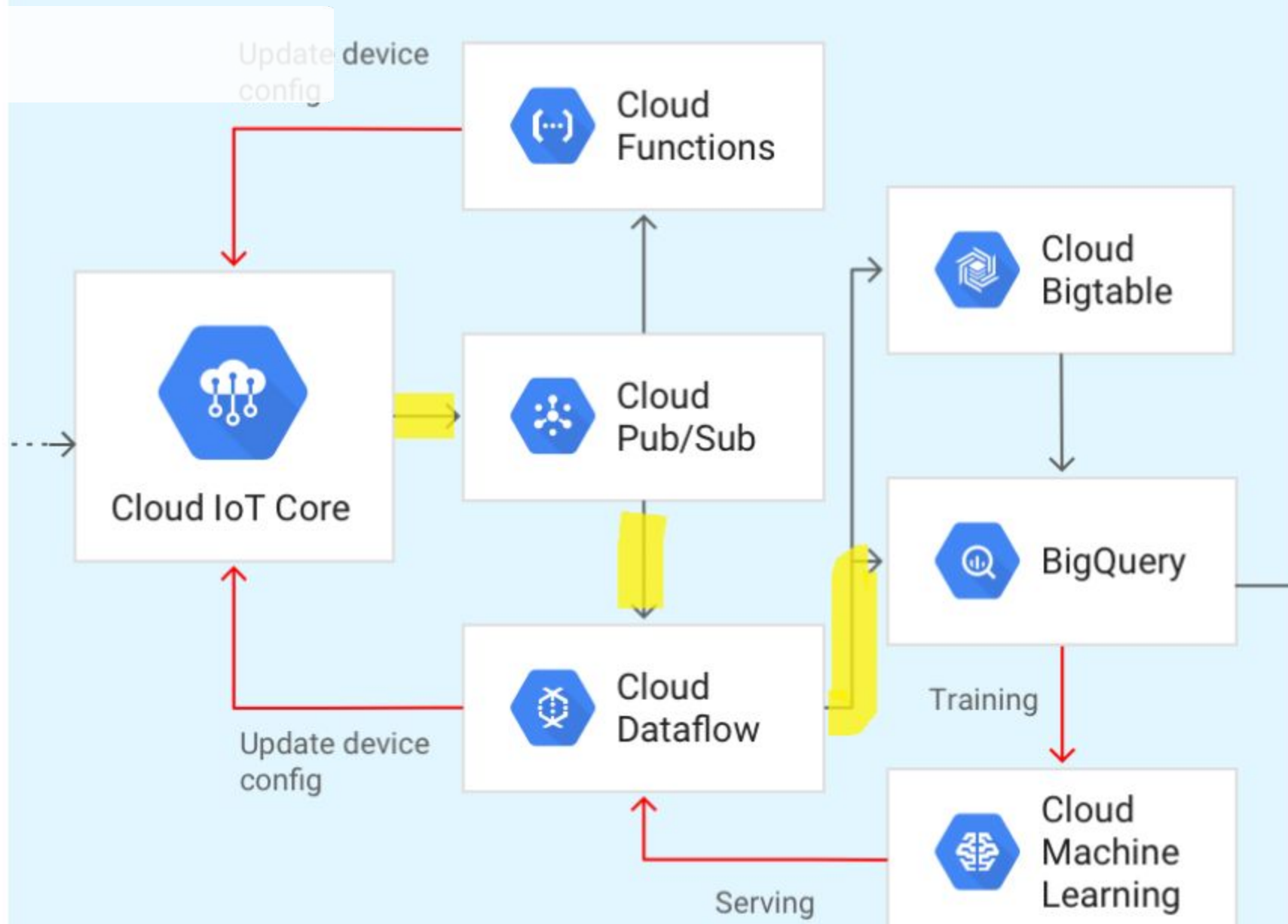
Process connected camera streams to detect objects, people, or quality defects on a line.



Vertex AI

Unified platform to deploy and scale your IoT machine learning models.

Data analytics in the cloud



Here is the holistic view. Note the unidirectional flow of data from devices to storage, and the feedback loops for device management.

IOT DATA PROCESSING PIPELINE



Ingest

Pub/Sub collects data via
MQTT/HTTP.



Process

Dataflow cleans and aggregates
data.



Store

Historical data saved in
BigQuery/Bigtable.

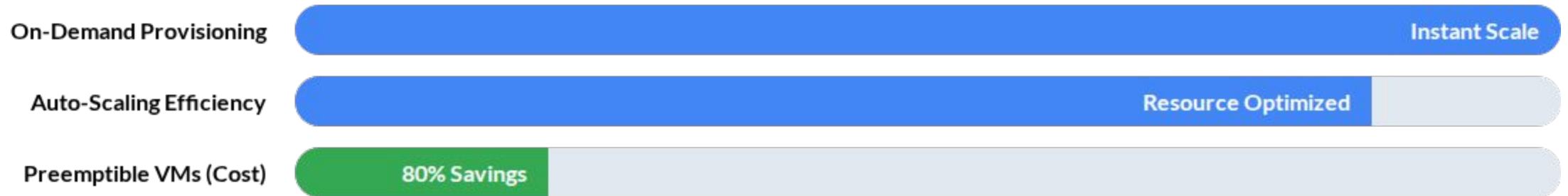


Visualize



Looker/Data Studio Dashboards.

PROVISIONING & LOAD BALANCING



Load Balancing: Essential for distributing millions of device connections across multiple ingestion nodes to prevent bottlenecks.

USE CASE: SMART CITIES

- ✓ **Traffic Optimization:** Connected sensors detect congestion and adjust light timings.
- ✓ **Waste Management:** Bins alert collection trucks only when full, saving fuel.
- ✓ **Environmental Monitoring:** Air quality sensors report pollution levels in real-time.

USE CASE: INDUSTRIAL IOT

- ✓ **Industry 4.0:** Complete digital twin of the factory floor in GCP.
- ✓ **Energy Management:** Monitoring machine power draw to optimize usage.
- ✓ **Asset Tracking:** Global GPS tracking of containers using GCP networking.



HEALTHCARE & AGRICULTURE IOT



Healthcare

Remote patient monitoring with wearables. Data is encrypted and HIPAA-compliant on GCP.



Agriculture

Soil moisture and weather sensors automate irrigation systems, increasing crop yield by 20%.



Smart Homes

Integrating millions of thermostats and lights with Google Home ecosystem on GCP infrastructure.

CHALLENGES IN CLOUD-IOT INTEGRATION

Security & Privacy

Encryption at rest and in transit is standard, but device-level compromise remains the biggest threat.

Latency & Bandwidth

Sending 4K video streams from 10,000 cameras to the cloud is expensive and slow. Need for Edge Computing.

Data Management

Filtering "Signal from Noise." Handling massive volumes of redundant data before it reaches expensive storage.

FUTURE: EDGE COMPUTING & AI

Edge TPU & AI

Moving machine learning to the device itself. Google's Edge TPU allows AI inference with millisecond latency.

Key Future Factors

- ✓ **5G Integration:** Massive throughput for connected vehicles.
- ✓ **Serverless Edge:** Functions running at the CDN level.
- ✓ **Autonomous IoT:** Devices making local decisions without cloud approval.

CONCLUSION

- ✓ GCP Functional Architecture provides a highly scalable and secure foundation.
- ✓ IoT services like Pub/Sub and BigQuery enable real-time insights from global device fleets.
- ✓ Future success depends on the synergy between Cloud, Edge, and AI.

References:

1. NIST Cloud Computing Definition (SP 800-145).
2. Google Cloud Documentation - IoT Solutions.
3. "Industrial IoT Architecture on GCP" - Research Journal 2024.